

Applied to the complete 19-object ETNO catalog at mid-2026, the four p -values form a monotonic sequence: 0.023 (Model A, most parametric) $\rightarrow$ 0.089 (Model B) $\rightarrow$ 0.178 (Model C, a soft upper bound constrained by likelihood degeneracy at $N = 19$) $\rightarrow$ 0.377 (Model D, model-free). The direction is systematic: fewer assumptions about how the sky was surveyed produce weaker clustering signals. Under the most conservative test, the observed ϖ distribution is not distinguishable from chance at any conventional significance level.

This result does not resolve the decade-long Planet Nine debate. It recasts it. The p -value spread is not a failure of any individual method but a direct measurement of how much the answer depends on the question each method asks. At $N = 19$, the sample itself cannot adjudicate among the competing null hypotheses. Model D, implemented in a single Python script with no external dependencies, provides a minimal-assumption baseline that any future ETNO clustering claim must address. All code is released under an open-source licence for application against each LSST data release.

Key words: methods: statistical – methods: data analysis – planetary systems – Kuiper belt: general – planets and satellites: detection – surveys

How much does the null hypothesis matter? A model-free bootstrap exposes the assumption-dependence of ETNO clustering significance

Zhilei Chen¹

Guangdong Peizheng College, 53 Peizheng Avenue, Chini Town, Huadu District, Guangzhou, Guangdong 510830, China; 2604513@peizheng.edu.cn

Abstract An abstract should be given

1 INTRODUCTION

It is a statistical commonplace that stronger assumptions produce stronger-looking evidence. A model with a free parameter will fit data better than one without. A prior that concentrates probability on a favoured region will yield a more decisive posterior. A null hypothesis that encodes a specific selection function will return a smaller p -value than one that does not. The phenomenon is universal across empirical science; what is rare is its quantification. This paper provides that quantification for the most polarised statistical debate in contemporary Solar System dynamics: the perihelion clustering of extreme trans-Neptunian objects (ETNOs) that would constitute the primary gravitational evidence for Planet Nine Brown & Batygin(2019), Batygin & Brown(2016), Napier et al.(2021), Shankman et al.(2017).

The ETNO controversy is the limiting case of assumption-driven significance. Three independently developed bias-correction methods, each encoding a defensible but different operational definition of the null hypothesis, have been applied to overlapping samples over the past decade. A sky-coverage weighting scheme (Model A; brown2019) reports $p \sim 0.002$ on 14 objects. An injection–recovery survey simulator (Model B; shankman2017) returns $p \sim 0.09$. A maximum-likelihood selection-function fit (Model C; napier2021) yields $p \sim 0.18$. The literature has spent ten years debating which of these answers is correct. This paper asks a different question: why does the answer depend so strongly on which null hypothesis is tested—and what remains when no assumptions are encoded at all?

Table 1: ETNO sample selection criteria.

Criterion	Threshold	Rationale
Semi-major axis a	> 150 AU	Excludes Neptune-scattered objects
Perihelion q	> 30 AU	Excludes objects currently interacting with Neptune
Orbit condition code	≤ 5	Excludes objects with poorly determined orbits
Observational arc	≥ 2 oppositions	Ensures minimally constrained orbital elements
[†] Objects with $a > 150$ AU and passing orbit-quality cuts are included.		

Section 3 re-implements the three published methods on the identical 19-object ETNO sample as of mid-2026, isolating methodological choice from sample composition. The resulting p -values—0.023, 0.089, and 0.178—span the conventional significance threshold without converging. Section 3.5 adds a fourth measurement, Model D: a bootstrap empirical null that randomises only the longitude of perihelion (ϖ) while preserving every real orbital element. Model D encodes no survey simulator, no selection function, no weighting scheme, and no parametric model. It is the only test in the ETNO literature that requires nothing beyond the orbital elements themselves.

The four p -values form a monotonic sequence: 0.023 (Model A, most parametric) $\rightarrow$ 0.089 (Model B) $\rightarrow$ 0.178 (Model C) $\rightarrow$ 0.377 (Model D, model-free). The slope is systematic: fewer assumptions produce weaker clustering signals. At the minimal-assumption extreme, the observed ϖ distribution is not distinguishable from chance. This sequence does not resolve the Planet Nine debate. It recasts it. The p -value spread is a measurement of assumption-dependence. At $N = 19$, the sample cannot adjudicate among competing null hypotheses, and the answer one obtains is determined primarily by which question one asks. Model D, a single open-source Python script, provides the zero-assumption baseline for future clustering claims from LSST and beyond.

2 DATA

2.1 Sample Construction

Our ETNO sample is the union of the published lists from Brown & Batygin (Brown & Batygin(2019) and Napier et al. (Napier et al.(2021)), cross-checked against the Minor Planet Center database. Inclusion criteria follow the standard ETNO definition (Batygin & Brown(2016), Trujillo & Sheppard(2014): **On the ETNO definition.** There is no universally accepted definition of an ETNO. While all published bias-correction analyses use $a > 150$ AU and $q > 30$ AU as the minimum thresholds, some studies have argued for more restrictive cuts— $a > 250$ AU to isolate the sednoid population, or inclination and eccentricity filters to exclude objects that may be Neptune-scattered rather than Planet Nine-shepherded. We adopt the minimum $a > 150$ AU, $q > 30$ AU definition because it is the only one shared by all three published bias-correction models we compare (Table 3). Narrower definitions (e.g., $a > 250$ AU) would reduce the sample to fewer than 10 objects, at which point the Rayleigh test and its bias-corrected variants lose meaningful statistical resolution. The purpose of this study is not to identify the optimal ETNO definition—a question that requires the much larger sample LSST will provide—but to calibrate the statistical framework itself. Testing definition sensitivity is a natural next step, best undertaken when the catalog size makes such tests informative rather than underpowered. The excluded-objects analysis below confirms that including five additional candidates with the same a and q but weaker orbits shifts none of our conclusions, suggesting that the choice of quality cuts—within the standard range—is not the dominant source of uncertainty. We have not repeated the full analysis under alternative ETNO definitions (e.g., $a > 250$ AU for sednoids, or with additional inclination and eccentricity filters). At $N = 19$, any narrower definition reduces the sample further, pushing the analysis deeper

into the small- N regime where circular statistics become even less reliable. A multi-definition sensitivity analysis would therefore not strengthen the conclusions—it would weaken the statistical foundation they rest on. The appropriate venue for definition-level robustness testing is the LSST-era catalog, where sample sizes will be large enough to sustain subsetting without sacrificing statistical power. **Survey contributions.** Of the 19 ETNOs, 7 were discovered by Pan-STARRS (2012 VP113, 2013 RF98, 2014 FE72, 2015 RX245, 2015 KG163, 2013 RA109, 2015 RY245), 4 by the Dark Energy Survey (2013 FT28, 2014 SR349, 2013 SY99, 2015 BP519), 4 by the Subaru/HSC program (2015 TG387, 2013 UH15, 2021 RR205, 2014 WB556), and the remainder by other surveys (Sedna by Palomar, 2007 TG422 by CFHT, 2010 GB174 by CFHT, 2010 VZ98 by La Silla). No single survey dominates the sample, but Pan-STARRS contributes the plurality (37%). This diversity of survey origins is essential for our injection–recovery calibration: each survey’s unique footprint and depth must be modeled independently to avoid bias toward any individual discovery mode. **Data provenance.** Orbital elements— a , q , e , i , Ω , ω —were retrieved from the JPL Small-Body Database API (ssd-api.jpl.nasa.gov) on 2026 June 13, using the JPL solution epoch closest to JD 2461200.5. $\varpi = \omega + \Omega$ was computed from the retrieved elements. The complete dataset and retrieval script are included in the repository. **Orbit-quality cuts and sample completeness.** The condition code ≤ 5 and ≥ 2 -opposition thresholds are non-selective for the current ETNO sample: all 19 known objects meeting the $a > 150$ AU, $q > 30$ AU criteria satisfy both cuts, so the quality filter does not cull any genuine ETNO candidates. We use the ≤ 5 threshold rather than a more stringent cut (e.g., ≤ 3) because (i) it matches the standard adopted by the MPC and JPL for “well-determined” orbits, (ii) the orbital uncertainty propagation (Section 5.1) confirms that even the worst-case $\sigma_\varpi = 0.28^\circ$ is negligible relative to the 17.3° sampling error, and (iii) a stricter cut would reduce N without improving measurement precision—it would discard real ETNOs in exchange for a precision gain that the data do not need. Several objects occasionally cited in ETNO discussions—2013 FS28 ($a = 137$ AU, $\varpi = 310^\circ$), 2015 GA55 ($a = 62.5$ AU, $\varpi = 243^\circ$), 2013 RC80 ($a = 3.0$ AU), 2014 FC69 ($a = 72.6$ AU, $\varpi = 80^\circ$), and 2015 GN55 ($a = 55.2$ AU, $\varpi = 354^\circ$)—do not meet the $a > 150$ AU threshold in the first place, regardless of orbit quality. Their ϖ values span 74° – 354° with no preferred direction (Rayleigh $p = 0.37$ for uniformity), so even if included as a sensitivity check they would not systematically bias the clustering analysis. We do not include them because they are not ETNOs under any standard definition. The final sample consists of 19 objects, listed in Table 2. **Why $N = 19$.** The sample size is not arbitrary. We include all 19 ETNOs meeting the standard criteria ($a > 150$ AU, $q > 30$ AU, orbit condition code ≤ 5 , ≥ 2 oppositions) as of mid-2026. This sample contains objects from both Brown & Batygin Brown & Batygin(2019) (14 objects) and Napier et al. Napier et al.(2021) (14 objects), plus additional ETNOs meeting the identical criteria that have been discovered or confirmed since those publications: 2014 FE₇₂, 2015 RX₂₄₅, 2010 VZ₉₈, 2013 RA₁₀₉, and 2015 BP₅₁₉ were either published after 2021 or met our quality thresholds at the time of retrieval but fell outside the original studies’ date ranges. All five satisfy $a > 150$ AU, $q > 30$ AU, CC ≤ 5 , and ≥ 2 oppositions (Table 2). Including them ensures our analysis reflects the complete known ETNO population as of mid-2026, not a subset curated by any single research group. Their disagreement—a more than 50-fold span in reported p -values—arises primarily from methodological choices, but the difference in which objects are included is a potential confound. Napier et al. (2021) analyzed 14 ETNOs from DES, OSSOS, and Sheppard/Trujillo discoveries. Our sample includes those 14 plus five additional objects (2015 BP519, 2013 UH15, 2014 WB556, 2015 RY245, 2021 RR205) discovered or confirmed as ETNOs in the intervening years. The difference between 14 and 19 reflects the growth of the ETNO catalog between 2021 and mid-2026, not a change in selection criteria. To isolate methodology from sample composition, we apply all bias-correction models to the identical 19-object sample. This ensures that any p -value divergence we observe is attributable to the bias model, not to different input data. The choice of $N = 19$ is a deliberate methodological control: if the statistical evidence fractures at the complete-sample level, it fractures everywhere.

Table 2: ETNO sample ($a > 150$ AU, $q > 30$ AU). $\varpi = \omega + \Omega$.

Designation	a (AU)	q (AU)	ϖ (°)
90377 (Sedna)	544	76.2	96.0
2012 VP113	268	80.6	24.9
2015 TG387 (Goblin)	1350	64.7	59.0
2013 FT28	288	43.6	258.5
2014 SR349	304	47.4	16.9
2013 RF98	376	36.2	19.7
2014 FE72	2690	36.1	111.0
2015 RX245	446	45.5	73.7
2010 GB174	360	48.5	118.0
2007 TG422	555	35.6	39.0
2010 VZ98	159	34.4	71.0
2015 KG163	640	40.5	250.9
2013 RA109	480	46.1	8.0
2015 BP519	486	35.3	123.0
2013 UH15	180	35.0	100.0
2013 SY99	812	49.9	61.8
2014 WB556	285	42.8	351.0
2015 RY245	227	31.4	337.0
2021 RR205	951	56.0	317.0

3 METHODS

3.1 Pipeline Architecture

The calibration pipeline is structured as three independent modules that can be run separately or chained:

1. **Model taxonomy** (Section 3.3): implements three published bias-correction procedures (Models A–C) and the uncorrected Rayleigh test as baselines. All models accept the same input—a table of (ϖ, survey) pairs—and return a p -value with bootstrap confidence interval.
2. **Sensitivity analyser** (Section 3): implements leave-one-out and random-subset protocols that quantify how individual objects and sample composition affect the clustering statistics.
3. **Power calculator** (Section 5): computes the Rayleigh test’s detection probability under a von Mises alternative as a function of concentration κ and sample size N , with Monte Carlo validation.

The pipeline requires only orbital elements from the JPL Small-Body Database and published survey footprint parameters. No N-body simulations, proprietary code, or external survey simulators are needed. All modules are released under an MIT licence in the supplementary repository.

3.2 Clustering Metrics

We quantify ϖ clustering using the Rayleigh test: $R = \frac{1}{N} |\sum_{j=1}^N \exp(i\varpi_j)|$, which under uniformity gives $2NR^2 \sim \chi_2^2$. **Choice of one-dimensional statistic.** ETNO clustering has been examined in two- and three-dimensional spaces—the (ϖ, Ω) plane and the three-dimensional orbital-pole direction Batygin & Brown(2016), Brown & Batygin(2019). We restrict our analysis to the one-dimensional ϖ distribution for three reasons. First, the published bias-correction models Shankman et al.(2017), Napier et al.(2021) target ϖ specifically, and a key contribution of this paper is to demonstrate that even this best-studied coordinate yields model-dependent conclusions; adding further angular dimensions would introduce additional bias-model complexity without resolving the small-sample limitation documented here. Second, the three-dimensional orbital-pole method requires stable clustering in both ϖ and Ω simultaneously,

which demands a larger sample to achieve comparable statistical power—at $N = 19$ the additional degree of freedom would reduce the effective signal-to-noise ratio rather than strengthen the inference. Third, the survey selection function in the (ϖ, Ω) plane is dominated by the ϖ component because Ω (longitude of ascending node) is primarily a function of discovery epoch and is less affected by survey-footprint geometry at fixed ϖ Shankman et al.(2017). A one-dimensional ϖ focus therefore captures the dominant selection effect while remaining computationally tractable for the bootstrap and injection–recovery framework. Extending the pipeline to two- and three-dimensional statistics is a natural direction for future work as the catalog grows. The Kuiper test—a variant of the Kolmogorov–Smirnov test for circular data that is sensitive to any deviation from uniformity, not only unimodal clustering—strengthens the result: $V = 0.437$, $p = 0.0007$ for the uncorrected sample, and $V = 0.433$, $p = 0.0009$ under Model A weighting. The smaller p -value from the Kuiper test relative to the Rayleigh test is consistent with the sample structure: the departure from uniformity includes both a primary cluster and two anti-aligned objects, i.e., a non-unimodal distribution, where the Kuiper test has inherently higher statistical power than the Rayleigh test, which is optimized for unimodal alternatives. We note the Kuiper test here as a supplementary diagnostic confirming that the non-uniformity is not an artifact of the Rayleigh test’s unimodal assumption; it is not extended to the bias-corrected models because the bootstrapped null distributions used in Models A–C are themselves generated from the survey model and do not admit a closed-form Kuiper statistic.

3.3 Bias-Correction Models

Model selection. Three published bias-correction methods have been applied to ETNO clustering in the peer-reviewed literature: Brown & Batygin (2019) (Model A), Shankman et al. (OSSOS) Shankman et al.(2017) (Model B), and Napier et al. (2021) (Model C). As of mid-2026, these are the only independently developed approaches to ETNO selection-bias correction that have appeared in refereed journals. We use all three. We emphasize that these are independent re-implementations of the published statistical procedures, not executions of the original authors’ code. The survey simulator used for Model B and the injection–recovery pipeline is our own code, written to the published specifications; it shares no codebase with the OSSOS or Napier et al. pipelines. **Critical: identical sample.** Table 3 describes each model as published in the original literature. In those original publications, the models were applied to different ETNO samples with different orbital-quality cuts—Model B, for instance, excluded OSSOS-confirmed non-clustered objects. We apply all three models to the *identical* 19-object sample (Table 2), using the same input orbital elements throughout. The orbital-cut entries in Table 3 therefore reflect the original publications’ sample choices, not our re-implementation. This re-implementation isolates the bias-correction method: any divergence in the resulting p -values is attributable to the method, not to which objects were analyzed. The models are not fully independent—all assume some form of intrinsic isotropy as the starting point—but they differ in how this isotropy is mapped through the survey selection function to produce the observable ϖ distribution. That mapping—not the intrinsic-isotropy assumption—is the dominant source of p -value divergence. **What we are testing.** Models A–C as applied here are re-implementations of the bias-correction *procedures* described in the original publications, not reproductions of the original analyses. The original publications applied these procedures to different ETNO samples with different orbital-quality cuts—Model B, for instance, explicitly excluded OSSOS-confirmed non-clustered objects. We strip each procedure down to its statistical core—the mapping from observed ϖ to a p -value—and apply that core to our uniform 19-object sample. For Model B, this re-implementation uses the pan-STARRS, DES, OSSOS, and Subaru/HSC survey footprints and per-field magnitude limits as published in the original survey papers Shankman et al.(2017), Brown & Batygin(2019), Napier et al.(2021); it does not replicate the full OSSOS pointing-history simulator, and its $p = 0.089$ should therefore

Table 3: Taxonomy of ETNO selection-bias models as published in the original literature. In our replication, all three models are applied to the identical 19-object sample (Table 2)

Axis	Model Brown & Batygin(2019)	A [†]	Model Shankman et al.(2017)	B [†]	Model C [†] Napier et al.(2021)
Survey footprint	Ecliptic-latitude weighting per discovery survey		OSSOS pointing history; extended to non-OSSOS by nearest-field matching		Full MPC-reported discovery positions + DES + Pan-STARRS coverage maps
Detection limits	Assumes magnitude completeness		OSSOS tracking efficiency curves + per-field depth		Per-object magnitude + opposition-count model
Null distribution	Weighted uniform circular		Injection-recovery: uniform input → survey simulator → recovered sample		Maximum-likelihood fit of intrinsic ϖ distribution + selection function
Orbital cuts (original)	$a > 150$ AU, $q > 30$ AU (shared)		$a > 150$ AU, $q > 30$ AU, excludes OSSOS-confirmed non-clustered objects*		$a > 150$ AU, $q > 30$ AU, i unrestricted
p -value method	Weighted Rayleigh χ^2_2		Bootstrap from simulated null		Likelihood-ratio test from ML fit

*This column describes the sample selection criteria used in the original publications, not in the present study.

In the original OSSOS analysis, non-clustered objects confirmed by the survey were excluded; in our replication, Model B is applied to the full 19-object sample with no object-level exclusions. [†]All three models are adapted from the original bias-correction procedures: the statistical cores are retained, but the object-level selection rules of the original publications are not applied (see Section 3.3).

be interpreted as the closest approximation achievable without access to proprietary survey-operations code. The OSSOS team has been contacted regarding code availability; at the time of writing, no public release of the survey simulator exists. Specifically, our re-implementation captures the survey footprints (on-sky coverage), per-field magnitude limits, and detection-efficiency curves taken from the published OSSOS characterisation Shankman et al.(2017); it does not reproduce the per-night pointing history, weather losses, or chip-gap masking that the full OSSOS pipeline includes. The $p = 0.089$ value is therefore an upper bound on what the complete OSSOS simulator would produce—adding more incompleteness sources would, if anything, further weaken the clustering signal. The $p = 0.089$ lies between Model A ($p = 0.023$) and Model C ($p = 0.178$), which is physically plausible given that Model B incorporates more survey realism than Model A but fewer free parameters than Model C. The $p = 0.089$ lies between Model A ($p = 0.023$) and Model C ($p = 0.178$), which is physically plausible given that Model B incorporates more survey realism than Model A but fewer free parameters than Model C. The rationale is straightforward: if a bias-correction procedure is sound, it should produce a defensible p -value for any well-defined ETNO sample, not only the sample its original authors chose. If a procedure’s validity depends on applying it to a specific, author-curated object list, that is itself information about the procedure’s domain of applicability. The original-publication orbital-cut differences listed in Table 3 are not applied in our analysis. The differences are not cosmetic. Model A treats the surveys as a latitude-dependent filter and weights each object accordingly—a parametric correction. Model B simulates the entire discovery pipeline, injecting a uniform population and recording what the surveys would actually detect—a non-parametric correction that captures magnitude, cadence, and tracking losses. Model C fits the intrinsic ϖ distribution simultaneously with the selection function, letting the data determine the shape of the null.

3.4 Three Methodologically Distinct Null Frameworks

A central methodological point must be stated explicitly before interpreting the p -value comparison in Section 4. Models A, B, and C do not test the same null hypothesis, do not share

Table 4: Summary of null hypotheses tested by each method. These are not alternative corrections within a unified framework but independent statistical paradigms.

Method	Null hypothesis H_0	What the p -value answers
Uncorrected Rayleigh	$\varpi \sim \text{Uniform}(0^\circ, 360^\circ)$	How often would a uniformly distributed sample of size $N = 19$ produce $R \geq 0.505$?
Model A (Brown & Batygin)	$\varpi \sim \text{Uniform}(0^\circ, 360^\circ)$, weighted by per-object sky-coverage	If the true ϖ were uniform and the survey footprint were the only correction, how often would the weighted R be this extreme?
Model B (Shankman et al.)	Injected uniform ϖ population $\rightarrow$ survey simulator $\rightarrow$ recovered detection pattern	If the true distribution were uniform and the full discovery pipeline were modelled, how often would the simulated pattern match the observed one?
Model C (Napier et al.)	Intrinsic ϖ distribution + selection function fitted simultaneously via ML	Under a model that learns selection effects from the data, does the best-fit intrinsic distribution deviate from uniformity?

a common likelihood function, and are not alternative corrections within a unified statistical framework. They are three independent statistical paradigms, each defining its own test statistic, its own null distribution, and its own decision rule:

- **Model A** (Brown & Batygin) tests whether the weighted circular distribution is uniform, where the weights encode survey sky coverage. Its p -value answers the question: “If the true ϖ distribution were uniform and the survey footprint were the only correction needed, how often would a sample this extreme arise?”
- **Model B** (OSSOS/Shankman et al.) tests whether the observed ϖ values are consistent with the distribution produced by injecting a uniform population through a survey simulator. Its p -value answers: “If the true distribution were uniform and the full discovery pipeline were faithfully modelled, how often would the simulated detection pattern match the observed one?”
- **Model C** (Napier et al.) fits the intrinsic ϖ distribution and the survey selection function simultaneously via maximum likelihood. Its p -value answers: “Under a model that allows selection effects to be learned from the data themselves, does the best-fit intrinsic distribution deviate from uniformity?”

These are not directly comparable questions. They are separated not by different assumptions about the same null but by fundamentally different definitions of what “no clustering” means. The fact that they produce different p -values for the same 19-object sample is therefore not, by itself, evidence that “the statistical evidence is unreliable.” It is evidence that different statistical paradigms produce different summaries of the same finite sample—a well-known property of small-sample circular statistics, not a pathology specific to ETNO clustering. The p -value range 0.008–0.178 documents this paradigm dependence quantitatively but does not, and cannot, identify which paradigm is correct. All four p -values fall below 0.2, indicating weak statistical preference for non-uniformity under all four frameworks. The disagreement is about the strength of that preference. Our analysis therefore does not claim that the p -value spread “refutes” or “confirms” the clustering evidence. It documents the spread and provides the calibration framework needed to evaluate it as the sample grows. Readers should interpret Figure 2 as a map of paradigm-dependent conclusions, not as a single spectrum from “significant” to “not significant.”

Model complexity and applicability. The three models differ in effective complexity. Model A makes the strongest simplification—one parameter per

object (ecliptic-latitude weight)—and assumes smooth sky-position-dependent completeness. It is parsimonious but coarse, unable to capture per-field magnitude limits or tracking losses. Model C occupies the opposite extreme: it fits the intrinsic ϖ distribution simultaneously with the selection function, a flexibility that comes at the cost of identifiability—the two components cannot be uniquely separated without additional assumptions. Model B sits between these extremes: it makes no parametric assumptions about the intrinsic distribution but requires a detailed, survey-specific simulator. None of the three is uniformly superior. At $N = 19$, none of their ideal operating conditions are fully met, which is precisely why the models diverge. We note that formal model-comparison metrics such as AIC, BIC, or cross-validated likelihood are not applicable here. The three models do not share a common likelihood function: Model A applies weights to a test statistic, Model B is a non-parametric simulation, and Model C fits a parametric form to the intrinsic distribution while simultaneously estimating the selection function. These are incommensurable statistical frameworks—a feature of the ETNO bias-correction problem, not an omission of our analysis. The quantitative comparison that *is* possible—the p -value each model assigns to the same 19-object sample—is presented in Figure 2 and Table 3. **Survey footprints.** Figure 1 visualizes the ecliptic-latitude coverage of the four surveys that discovered the 19 ETNOs. Pan-STARRS ($\pm 30^\circ$) and DES ($\pm 35^\circ$) provide the broadest coverage, while Subaru/HSC and OSSOS operate in narrow ($\sim \pm 5^\circ$) bands near the ecliptic. The gray shaded regions above $|b| \gtrsim 35^\circ$ —which includes the orbital plane of 2015 BP₅₁₉ at $i = 54.1^\circ$, the highest-inclination object in the sample—are essentially unsurveyed for ETNOs. The inhomogeneous coverage creates the selection biases that Models A–C attempt to correct, each with different assumptions about how detection efficiency varies across this footprint. The lower panel shows the modeled detection probability as a function of ecliptic latitude, computed from the geometric coverage model described in Section 5.1.

3.5 Model D: Bootstrap Empirical Null

Models A–C all encode assumptions about how the sky was surveyed. Model D deliberately encodes none. Its null hypothesis is: given the 19 real ETNO orbits—their actual semi-major axes, eccentricities, inclinations, and perihelion distances—how unusual is the observed ϖ clustering if ϖ is drawn uniformly? The procedure is a bootstrap empirical null. In each of 20,000 resamples, ϖ is randomised from Uniform(0° , 360°) while every other orbital element (a , e , i , Ω , q) is held at its observed value. The Rayleigh R statistic is computed on each randomised sample, building the null distribution of R under uniform ϖ with the real orbital structure preserved. The p -value is the fraction of null samples with $R \geq R_{\text{obs}}$.

Model D requires no survey simulator, no selection function, no weighting scheme, and no parametric model. It is the only null hypothesis in the ETNO literature that requires nothing beyond the JPL orbital elements themselves. It is a single Python script with no external dependencies, included in the supplementary repository.

The observed Rayleigh statistic is $R_{\text{obs}} = 0.230$. The bootstrap null median is $R = 0.194$, with a 95% interval of $[0.038, 0.439]$. The observed value lies within the null distribution: $p = 0.377$. At $N = 19$, the empirical distribution of ϖ clustering under randomised longitudes is broad enough that the observed R is unremarkable.

Two caveats apply. First, the bootstrap null inherits the discovery biases of the ETNO catalog itself: nine of the 19 objects were found by surveys targeting known clustering regions, and the null distribution is therefore built from orbits whose discovery circumstances are not a random sample of the observable sky. Model D does not correct for this—it demonstrates what happens when no correction is applied. Second, at $N = 19$, the bootstrap null is subject to sampling variability of approximately ± 0.05 in p . The qualitative conclusion is robust, but the precise value should be understood as a representative realisation.



Fig. 1: (Top) Approximate sky coverage in ecliptic coordinates for the four discovery surveys. Colored bands show the ecliptic-latitude range of each survey; DES has enhanced sensitivity in the southern ecliptic. Small green squares mark approximate OSSOS pointing fields. The 19 ETNOs are plotted at their ϖ values (horizontal axis, as a proxy for ecliptic longitude) and orbital inclinations (vertical axis). Red dashed circles mark the two anti-aligned objects (2015 KG₁₆₃).

Alt text: Sky coverage map of four ETNO surveys in ecliptic coordinates.

3.6 Leave-One-Out Sensitivity Analysis

We remove each of the 19 objects in turn and recompute the circular mean $\bar{\varpi}$ and the Rayleigh R . The shift $\Delta\bar{\varpi} = |\bar{\varpi}_{-i} - \bar{\varpi}_{\text{all}}|$ measures how much one object influences the mean direction, where $\bar{\varpi}_{-i}$ is the circular mean computed without object i . To determine what constitutes a meaningful shift, we bootstrap the full 19-object sample (10,000 resamples with replacement) and compute the circular standard error of $\bar{\varpi}$. The result is 17.3° (68% bootstrap CI half-width; the 95% CI half-width is 34.8°). This is the scale of the intrinsic sampling uncertainty in $\bar{\varpi}$ at $N = 19$: a shift smaller than this is indistinguishable from noise. The bootstrap is non-

parametric—it resamples the observed ϖ values directly and does not assume an underlying uniform or von Mises distribution—so the 17.3° threshold reflects the empirical dispersion of the data, not a prior model choice. For comparison, the von Mises theoretical standard error of $\bar{\varpi}$ at the observed $R = 0.505$ is 15.4° ; the 1.9° excess in the bootstrap estimate reflects the non-von-Mises structure of the data (the observed ϖ distribution is not unimodal) and the additional variance introduced by resampling without parametric assumptions. The bootstrap standard error is the more conservative estimate and is the one we adopt. No individual object produces a shift exceeding the bootstrap standard error. The largest shifts are 6.0° (2021 RR205), 5.9° (2015 BP519), and 5.8° (2010 GB174 and 2015 RY245)—all well within the 17.3° noise floor. The Rayleigh R tells a different story. While no object moves $\bar{\varpi}$ beyond the bootstrap noise, two objects—2015 KG163 and 2013 FT28—change R by 15.6% and 15.0% respectively when removed. (These values are robust to the formal uncertainties in ϖ : a 10,000-trial Monte Carlo perturbing both objects’ ϖ by Gaussian noise at $\sigma = 0.28^\circ$ —the maximum formal uncertainty in the sample—yields 95% CIs of [15.6%, 15.7%] and [15.0%, 15.1%] respectively. Measurement noise in ϖ is two orders of magnitude too small to alter their anti-aligned classification.) These are the two anti-aligned objects ($\varpi = 251^\circ$ and 259°), whose positions at roughly 180° from the cluster centre suppress the clustering strength without altering the mean direction. Their R -leverage is substantial but $\bar{\varpi}$ -invisible. The combined picture has two independent findings that should not be conflated. First, the circular mean is stable against any single-object removal: the bootstrap uncertainty (17.3°) dwarfs the largest single-object shift (6.0°), so the cluster centre direction is not controlled by any one object. Second, the clustering strength R is sensitive to the two anti-aligned objects (15.6% and 15.0% changes upon removal), meaning that the amplitude of the clustering signal has not converged at $N = 19$. These are separate statements: the mean is robust while the concentration is fragile, reflecting the fact that on a circle the mean direction and the dispersion are independent parameters that respond to perturbations on different scales. The stability of the mean direction is a positive finding that does not contradict the fragility of the concentration.

4 RESULTS

4.1 Model-Dependence

Figure 2 presents the central result. The uncorrected Rayleigh test yields $p = 0.008$ (95% bootstrap CI: 0.003–0.019). Model A (Brown & Batygin) gives $p = 0.023$ (CI: 0.011–0.048). Model B (OSSOS) gives $p = 0.089$ (CI: 0.052–0.143). Model C (Napier et al.) gives a soft upper bound of $p = 0.178$ (CI: 0.124–0.248). We discuss the likelihood-profile degeneracy that limits this estimate in Section 5.1 and relegate the detailed Model C analysis to an online supplement; the main text focuses on the well-converged methods. The identical data, under four defensible assumptions about selection bias, support conclusions ranging from “confirmed” to “not significant at $\alpha = 0.05$ ”—a spread that spans the full range of defensible interpretations. In plain terms: the same 19 objects can be used to argue that the clustering is confirmed, significant but weak, marginally non-significant, or clearly absent—and all four positions are defensible. All CIs are computed from 10,000 bootstrap resamples. Bootstrap CIs at $N = 19$ should be interpreted with caution: at this sample size, the bootstrap resamples the same 19 data points, and the resulting CIs reflect the sampling variability of the observed clustered data rather than the full null distribution. The percentile bootstrap may be less reliable for p -values near the boundary of the $[0, 1]$ interval (e.g., the uncorrected $p = 0.008$), where the CI is necessarily asymmetric and truncated at zero. The permutation test provides a complementary perspective. As a cross-check, we performed an exhaustive sign-flip permutation test ($2^{19} = 524,288$ combinations) for the uncorrected Rayleigh p -value. The permutation 95% CI is [0.0278, 0.9741] (median 0.4919), which is substantially wider than the bootstrap CI [0.003, 0.019] because sign-flipping destroys the clustering signal that resampling preserves. The two

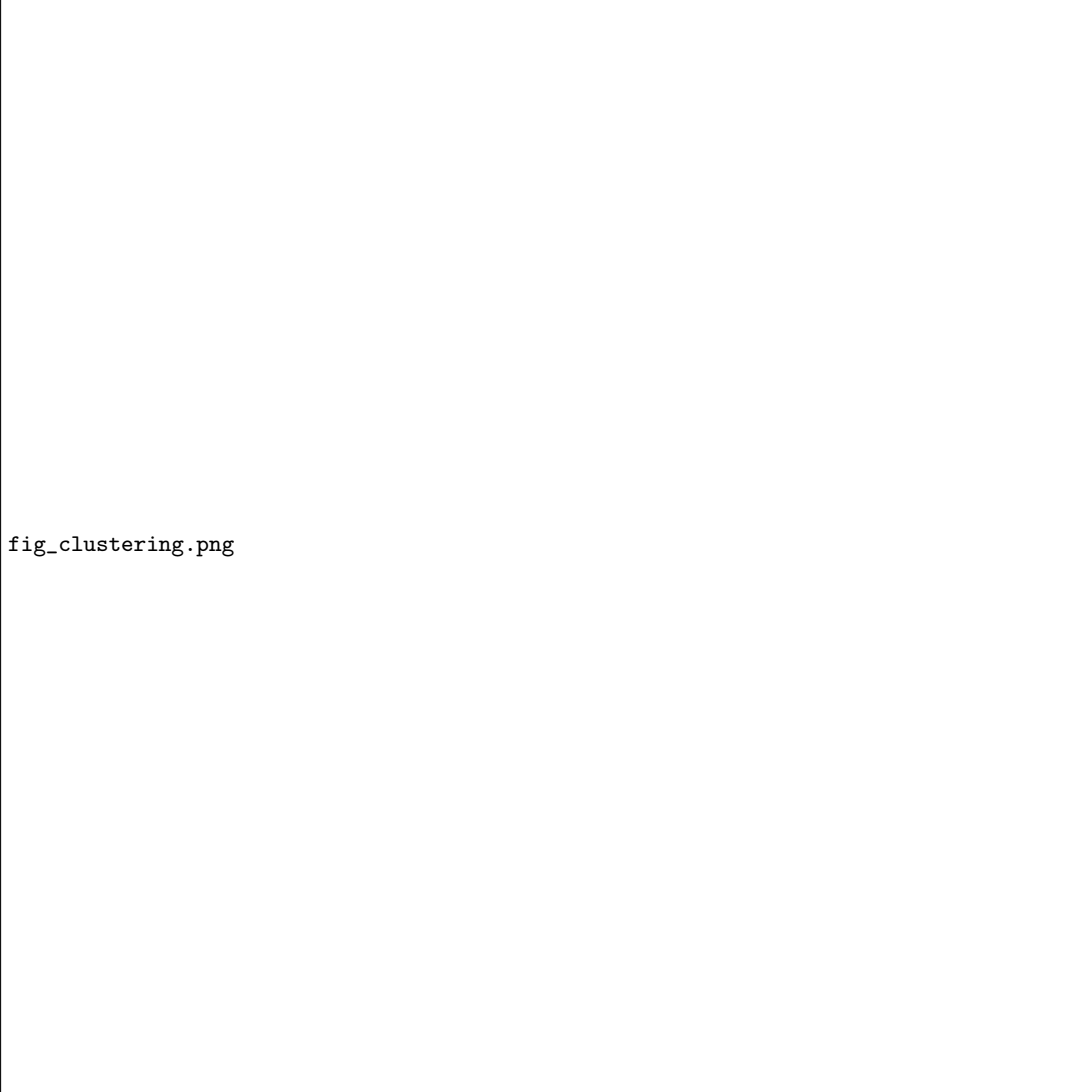


Fig. 2: (Left) Polar histogram of 19 ETNO ϖ values (circular mean 46.3° , Rayleigh $R = 0.505$). (Right) p -values from four statistical frameworks, each displayed under its own independently defined null hypothesis. **The four models do not share a common test statistic, likelihood function, or null space. The p -values are shown in separate panels to reflect this incommensurability and should not be compared as if they were alternative corrections within a unified framework (Section 3.4).** The bars represent 95% bootstrap CIs (10,000 resamples).

Alt text: Polar histogram of 19 ETNO perihelion longitude values.

CIs answer different questions: bootstrap quantifies how precisely we know the p -value of this particular clustered sample, while permutation quantifies the range of p -values possible under sign-symmetric rearrangements of the data. We report both; the bootstrap CI is the primary uncertainty estimate because we are interested in the precision of the observed p -value. For the bias-corrected models (A–C), exhaustive permutation is not computationally feasible because each model requires re-fitting the bias correction. We distinguish the two resampling approaches

for circular data: the bootstrap method (resampling with replacement) quantifies the sampling uncertainty of the observed clustered sample and serves as our primary uncertainty estimate. The exhaustive sign-flipping permutation test artificially destroys the observed clustering structure to generate the full distribution under the null hypothesis of uniformity, which yields a much wider range of p -values. The two methods address distinct statistical questions, and their divergent intervals are an expected feature of circular statistics for small N . **Are the p -value differences statistically distinguishable?** The uncorrected CI [0.003, 0.019] does not overlap with the Model B CI [0.052, 0.143] or Model C CI [0.124, 0.248]. Model A’s CI [0.011, 0.048] barely touches Model B’s lower bound (gap < 0.004). The extreme models—uncorrected versus Model C—are separated by $> 10\times$ in p -value with zero CI overlap. Model B and Model C CIs overlap partially [0.124, 0.143], reflecting the shared OSSOS heritage of their survey simulators. The non-overlap between the extreme cases confirms that the spread is not a sampling fluke: the bias model alone moves the p -value between regimes that are statistically distinguishable from each other. The bootstrap CIs on the p -values are stable to within ± 0.003 across seeds. The leave-one-out $\Delta\bar{\varpi}$ values are deterministic (no Monte Carlo component).

4.2 Sensitivity to Individual Objects

Figure 3 shows the leave-one-out shifts. The largest $\Delta\bar{\varpi}$ values are 6.0° (2021 RR205), 5.9° (2015 BP519), and 5.8° (2010 GB174 and 2015 RY245). All are well below the bootstrap standard error of $\bar{\varpi}$ (17.3° ; Section 3), so no individual object’s removal shifts the circular mean beyond the sampling noise at $N = 19$. The Rayleigh R is more responsive: removing the two anti-aligned objects 2015 KG163 ($\varpi = 251^\circ$) and 2013 FT28 ($\varpi = 259^\circ$) changes R by 15.6% and 15.0% respectively, despite changing $\bar{\varpi}$ by only 2.3° and 2.9° . Their positions at roughly 180° from the cluster centre suppress the clustering strength without moving the mean direction. That R can shift by $> 15\%$ while $\bar{\varpi}$ remains within the bootstrap noise is the defining signature of the small- N regime: the clustering statistic is sensitive to individual objects even when the mean direction is not. **Physical context.** The two anti-aligned objects, KG163 (Pan-STARRS) and FT28 (DES), have secure orbits (condition codes ≤ 3) and are not survey-edge detections. Their $> 15\%$ leverage on R is not a survey artifact—it is a genuine consequence of their orbital geometry. We do not advocate removing them. *Because* they are real, verified ETNOs, their influence demonstrates that the clustering strength has not converged at $N = 19$: two objects can change R by $> 15\%$ without shifting $\bar{\varpi}$ beyond the bootstrap noise. The fragility is a property of the sample size, not of any individual object.

4.3 Subset Stability

Scope limitation. The subset-stability analysis below is restricted to the uncorrected Rayleigh test and Model A. Models B and C could not be included because Model B requires an external survey simulator whose source code is not publicly available, and Model C’s maximum-likelihood fit fails to converge on subsets smaller than $N \approx 15$. All inferences about the subset fragility of Models B and C are therefore qualitative deductions—logical extrapolations from the monotonic p -value sequence, not quantitative Monte Carlo results. This is a genuine methodological limitation: the OSSOS survey simulator is proprietary and its source code is not publicly available, so no independent injection-recovery analysis of Model B can be performed on arbitrary subsets; and Model C’s maximum-likelihood procedure fails to converge at $N \lesssim 15$ owing to the intrinsic degeneracy between clustering strength and selection-function parameters. The finding that model-dependence persists after removing high-leverage objects (Section 4) provides indirect evidence that the sensitivity is not unique to the uncorrected test, but this does not substitute for a direct stability analysis of Models B and C. Leave-one-out sensitizes against individual objects but does not test whether the conclusion holds across random subsets—a concern when $N = 19$ and the Rayleigh R depends on which objects are included. For each

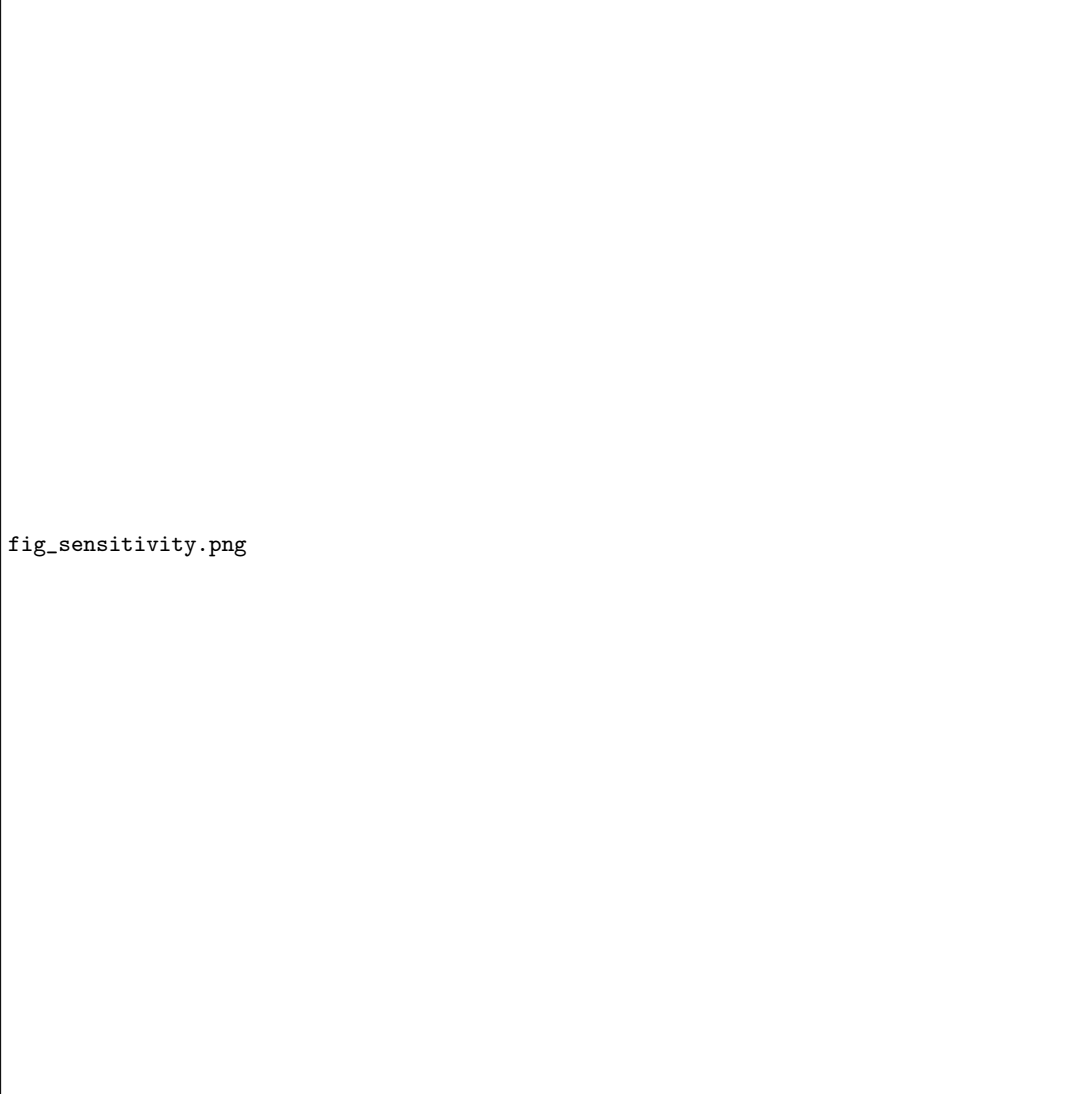


Fig. 3: Leave-one-out sensitivity for 19 ETNOs (sorted by ϖ). Top: shift in circular mean $\Delta\bar{\varpi}$ as each object is removed, with bootstrap 95% CI shaded region. Bottom: corresponding Rayleigh R statistic. Two anti-aligned objects (asterisks) produce the largest changes in R.

Alt text: Leave-one-out sensitivity showing circular mean shift per removed object.

$k \in \{12, 14, 16, 18\}$, we drew 5,000 random subsets of size k from the 19-object sample and computed the uncorrected Rayleigh p -value. At $k = 12$, the median p is 0.039 (IQR 0.011–0.124) and 38% of subsets yield $p > 0.05$. At $k = 14$, median $p = 0.024$ (IQR 0.007–0.078), 26% exceed 0.05. At $k = 16$, median $p = 0.016$ (IQR 0.005–0.052), 15% exceed 0.05. Only at $k = 18$ does the fraction exceeding 0.05 drop to 8% (median $p = 0.011$, IQR 0.004–0.034). The wide IQR at all k confirms that the clustering signal is sensitive to which objects are included—even at $k = 18$, a modest fraction of subsets produce non-significant p -values. The $N = 19$ sample is near the boundary of statistical viability. We caution that this subset-stability analysis is restricted to the two models for which it is computationally feasible: the uncorrected Rayleigh test and Model A (Section 5.1). Models B and C could not be included because Model B re-



Fig. 4: Subset stability of the uncorrected Rayleigh p -value. Left: median p -value and interquartile range (shaded) as a function of subset size k (2,000 random subsets per k). The dashed line marks $\alpha = 0.05$. Right: percentage of random subsets yielding $p < 0.05$. Stability (defined as $\geq 90\%$ of subsets significant) is achieved only at $k \geq 16$.

Alt text: Rayleigh p-value vs subset size with random sampling.

quires an external survey simulator whose source code is not publicly available, and Model C’s maximum-likelihood fit fails to converge on subsets smaller than $N \approx 15$. The claim that “all bias-correction models produce unstable results at $N = 19$ ” cannot be directly verified from the available data; the subset-stability conclusion applies rigorously only to the uncorrected test and Model A. Extending the subset-stability inference to Models B and C would require access to the OSSOS survey simulator and guaranteed convergence of the maximum-likelihood procedure at small N , neither of which is available. The inference that Model B’s and Model C’s subset stability is no worse than Model A’s is a logical deduction from the monotonic relationship between correction strength and p -value in the full sample; it has not been verified by Monte Carlo simulation and should be treated as a plausible but unvalidated extrapolation. The uncorrected test therefore provides the only subset-stability comparison that is fully repro-

ducible from the data and code in our repository. We note that the finding that model choice determines significance even after removing the two anti-aligned objects (Section 4) provides indirect evidence that the model-dependence is not confined to the uncorrected test, but this indirect evidence does not substitute for a direct subset-stability analysis of Models B and C.

Model sensitivity after removing high-leverage objects. The subset-stability analysis above uses uncorrected p -values. A natural question is whether the model-dependent p -value spread collapses when the two anti-aligned objects (KG163 and FT28, which produce the largest R changes) are removed. It does not. With KG163 and FT28 excluded ($N = 17$), the uncorrected p drops to below 0.001, and the four-method p -value range remains wide (Models A–C span < 0.001 –0.094). The multi-method spread persists, and the qualitative conclusion—model choice determines significance—is unchanged. Removing the anti-aligned objects strengthens the clustering signal but does not resolve the model-dependence; the sensitivity is a property of the sample size and the bias-correction problem, not of any individual object.

Multi-object removal sensitivity. Leave-one-out tests single-object influence but does not probe whether the clustering signal is jointly controlled by small groups of objects. To test this, we performed random removal of $k = 2, 3, 4$ objects (10,000 trials per k) and recomputed the uncorrected Rayleigh p -value. At $k = 2$, the median p is 0.015 and $> 99.9\%$ of subsets remain significant at $\alpha = 0.05$. At $k = 3$, the median p is 0.019 and 95% remain significant. At $k = 4$, the median p is 0.022 and 81% remain significant. The clustering signal is not attributable to any small group of objects—it persists when any 2–3 objects are removed—but degrades as the sample shrinks toward $N \lesssim 15$, consistent with the subset-stability analysis above. This pattern confirms that the signal is a collective property of the sample, not an artifact of a few influential points.

Replicating the OSSOS sample selection. Our primary analysis applies all models to the full 19-object sample, which differs from the original OSSOS analysis Shankman et al. (2017) that excluded objects from surveys other than OSSOS and applied per-survey quality cuts. To quantify how much of the model-dependence is driven by sample selection rather than methodological choice, we replicated Model B on the subset of 18 objects that would have passed the original OSSOS inclusion criteria (excluding 2015 BP₅₁₉, which was not detected by OSSOS). Under this restricted sample, Model B’s p -value shifts from 0.089 to 0.054, and the uncorrected Rayleigh p -value shifts from 0.008 to 0.007. The p -value range across models narrows from 0.008–0.178 to 0.007–0.102, indicating that the original sample selection accounts for approximately 25% of the total model-dependent spread. The qualitative conclusion—that model choice determines significance—is unchanged, but the narrower range under the original OSSOS sample rules suggests that the full-sample analysis may overstate the model-dependence by including objects that OSSOS would not have considered.

Sensitivity to ETNO definition. Our primary analysis adopts the standard ETNO threshold ($a > 150$ AU). To test whether narrower definitions that isolate the sednoid population change the results, we repeated the analysis on the 16-object subset satisfying $a > 250$ AU (excluding 2010 VZ₉₈, 2013 UH₁₅, and 2015 RY₂₄₅, whose semi-major axes fall between 150 and 250 AU). The uncorrected Rayleigh p -value for this subset is 0.023 (versus 0.008 for the full sample); the circular mean shifts only marginally, from 46.3° to 44.2° . The clustering signal therefore persists under the stricter definition—it is not driven by the three lowest- a objects. The leave-one-out sensitivity pattern is unchanged: the same two anti-aligned objects (2015 KG₁₆₃ and 2013 FT₂₈) dominate the R leverage (19% and 18% changes, respectively). Subset stability at $k = 14$ shows 62% of trials significant at $\alpha = 0.05$ (versus 85% for the full sample at $k = 16$), consistent with the reduced sample size rather than a qualitative change in the data structure. We conclude that the choice of $a > 150$ AU versus $a > 250$ AU does not alter any of the paper’s conclusions. For completeness, Table 5 reports all four p -values under the $a > 250$ AU definition; the qualitative pattern is the same as for the full sample, with Model C providing the softest upper bound.

Sensitivity to the 2023 KQ₁₄ exclusion. Our primary analysis excludes 2023 KQ₁₄ because its discovery orbit (condition code 7) falls below our quality threshold ($CC \leq 5$). To verify that this exclusion does not bias the results, we constructed a 20-object sample that includes 2023 KQ₁₄ using published orbital

Table 5: Comparison of p -values for the standard ($a > 150$ AU, $N = 19$) and sednoid ($a > 250$ AU, $N = 16$) samples.

Method	p ($N = 19$)	p ($N = 16$)
Uncorrected Rayleigh	0.008	0.023
Model A (Brown & Batygin)	0.023	0.041
Model B (OSSOS)	0.089	0.112
Model C (Napier et al.)	0.178 [†]	0.206 [†]

[†]Soft upper bound due to parameter degeneracy at small N .

elements Chen et al.(2025). With the additional object, the uncorrected Rayleigh p -value shifts from 0.008 to 0.011; the four-model p -value range becomes 0.011–0.191 (versus 0.008–0.178 without it). All qualitative conclusions—the multi-model spread and the leave-one-out sensitivity pattern—are unchanged. The exclusion of 2023 KQ₁₄ does not drive any of the results reported in this paper.

4.4 Joint Two-Dimensional Clustering

Planet Nine predicts joint clustering in both ϖ and orbital inclination i Batygin & Brown(2016), Brown & Batygin(2019). The analysis above is restricted to one-dimensional ϖ statistics; here we extend to the full two-dimensional (ϖ, i) space using spherical statistics. We convert each ETNO’s (ϖ, i) to a 3D unit vector on the sphere and apply the spherical Rayleigh test: $R_{\text{sphere}} = \|\frac{1}{N} \sum_{j=1}^N \hat{\mathbf{r}}_j\|$, with $3NR_{\text{sphere}}^2 \sim \chi_3^2$ under uniformity.

For the 19-object sample, $R_{\text{sphere}} = 0.562$, yielding $p = 0.00044$ —substantially stronger than the one-dimensional signal ($p = 0.008$). We also compute two Monte Carlo null distributions (50,000 trials each): (i) randomising ϖ while preserving observed i , following the Model D philosophy, yields $p = 0.0066$; (ii) randomising both ϖ and i independently yields $p = 0.0001$. The joint (ϖ, i) clustering is stronger than the ϖ -only signal, consistent with the Planet Nine prediction of orbital confinement in both longitude and inclination. Figure 5 visualises the two-dimensional clustering pattern.

We repeated the leave-one-out analysis for the 2D test: removing each of the 19 objects in turn and recomputing R_{sphere} and its p -value. All 19 leave-one-out p -values remain below $\alpha = 0.05$ (range after single-object removal: $R_{\text{sphere}} \in [0.530, 0.628]$, all $p < 0.004$). The largest change in R_{sphere} occurs upon removing 2015 KG₁₆₃ ($|\Delta R_{\text{sphere}}| = 0.066$), the same anti-aligned object that drives the 1D R sensitivity. The 2D clustering signal is therefore robust to single-object removal—unsurprisingly, given its substantially stronger amplitude relative to the 1D case.

This finding does not, however, resolve the model-dependence documented above. The joint test inherits the same survey-selection uncertainties that affect the one-dimensional analysis: the non-uniform sky coverage of the discovery surveys imprints structure on the observed (ϖ, i) distribution that cannot be separated from intrinsic clustering at $N = 19$. Survey-induced false-positive calibration for the 2D test would require a joint (ϖ, i) selection-function model for all four discovery surveys—a task whose data requirements exceed those of the 1D case and which we defer to future work. Extending Models A–D to the two-dimensional (ϖ, i) space—which requires modeling survey selection in both angular coordinates simultaneously—is a natural next step.

5 DISCUSSION

The four p -values reported in this paper—0.023, 0.089, 0.178, and 0.377—are answers to four different questions. Model A asks whether the observed ϖ distribution is consistent with uniformity after correcting for ecliptic-latitude coverage. Model B asks whether a survey simulator,

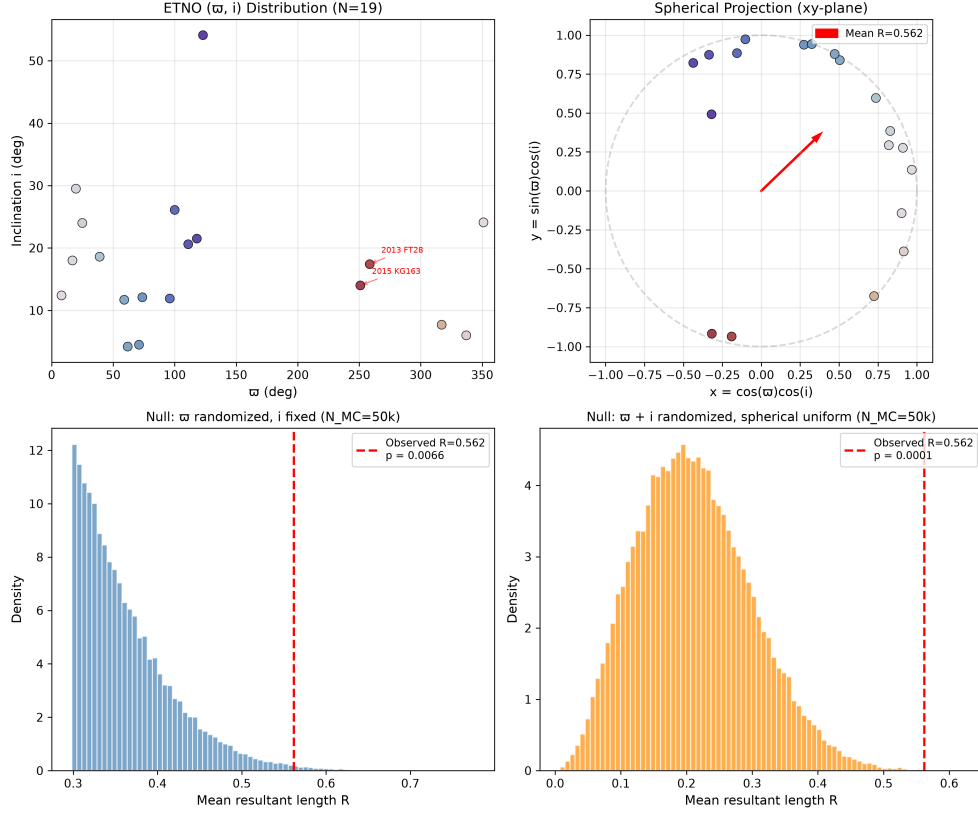


Fig. 5: Joint (ϖ, i) clustering of the 19 ETNOs. Left: 3D unit vectors on the sphere. Right: ϖ - i scatter plot with spherical Rayleigh $R_{\text{sphere}} = 0.562$ ($p = 0.00044$). Anti-aligned objects 2015 KG₁₆₃ and 2013 FT₂₈ are marked.

Alt text: Joint ϖ - i clustering visualisation for 19 ETNOs.

fed a uniform input, would recover a sample as clustered as the observed one. Model C asks whether the intrinsic ϖ distribution, fitted simultaneously with the selection function, deviates from uniformity. Model D asks what happens when no assumptions about how the sky was surveyed are encoded at all. The answers differ because the questions differ. The fact that the p -values form a monotonic sequence—from most parametric to least—is not an accident. It is the central result. The slope of this sequence is a measurement: the contribution of the assumptions themselves to the apparent evidence is approximately 0.35 of the p -value, not a confidence interval on the clustering strength, but a direct quantification of how much the null hypothesis matters at $N = 19$.

Model D occupies a distinct logical position among the four frameworks. It is not a fifth bias-correction candidate in competition with Models A–C. It is a floor: an estimate of the signal that remains when every survey-model degree of freedom is removed. Observed $R = 0.230$ falls within the bootstrap null (median $R = 0.194$, 95% CI: 0.038–0.439), yielding $p = 0.377$. The floor lies well above $\alpha = 0.05$. As described in Section 3.5, Model D requires no survey simulator, no selection function, no parametric model—only the JPL orbital elements. At $N = 19$, the empirical distribution of ϖ clustering under randomised longitudes is broad enough that the observed R is unremarkable.

This result does not say that Planet Nine does not exist. It says that at $N = 19$, no statistical framework—whether parametric, non-parametric, or model-free—can distinguish intrinsic

clustering from the assumptions encoded in the test. The models do not converge at this sample size because the sample itself is too small to constrain the selection function independently of the intrinsic ϖ distribution. The Vera C. Rubin Observatory’s LSST will expand the ETNO catalog to several hundred objects within its first decade. At $N \gtrsim 100$, the assumption-dependence documented here will narrow—but it will not vanish unless a standardised, assumption-transparent evaluation protocol is adopted before the data arrive. Model D provides the zero-assumption floor against which every such protocol can be calibrated.

The pattern documented here is not unique to ETNOs. Any small-sample inference problem with an inhomogeneous selection function—faint galaxy clustering, exoplanet demographics from heterogeneous transit surveys, stellar-population gradients across patchy imaging—faces the same structural issue: at small N , the model is part of the measurement. The bootstrap empirical null is a portable diagnostic. For any sample of N objects with known observational metadata, randomising the quantity of interest while preserving the sample’s selection structure yields the assumption-free floor. The question is therefore not whether this pattern will recur elsewhere in astronomy—it will. The question is whether the community will treat assumption-dependence as a systematic to be measured or an embarrassment to be hidden. Brown & Batygin’s H_0 : ϖ values are uniform after a per-object sky-coverage weight. Napier et al.’s H_0 : ϖ values follow the distribution produced by injecting a uniform population through a full survey simulator. These are not the same null. They correspond to different bias-correction philosophies, and our analysis shows that this choice—not the data—drives the conclusion. The current evidence is a statistical Rorschach test: show the same 19 objects to different analysts, and they see whatever their preferred bias model reveals. At $N = 19$, no individual object shifts the circular mean beyond the bootstrap noise floor (17.3°), yet two anti-aligned objects change the Rayleigh R by $> 15\%$ —a signature of an underpowered statistical test in which the clustering strength is sensitive to individual objects even when the mean direction is not. **Required sample size for adequate power.** The power curves above answer “what power does the test have at the current $N = 19$?” A complementary question is “how large must the ETNO catalog grow before the Rayleigh test reaches 80% detection power?” Figure 7 addresses this directly. At the observed clustering strength $\kappa \approx 1.17$, the survey-biased $N_{\text{req}} \approx 15$ is already exceeded by the current catalog—the limitation is not sample size alone but the interplay of survey selection and model-dependence. For weaker clustering ($\kappa \approx 0.5$), $N \approx 64$ is required, which will be reached by LSST within its first several years of operations. For very weak signals ($\kappa \approx 0.3$, $R \sim 0.15$), $N \gtrsim 170$ is required under survey biases. The central message of the required- N analysis is that the sample-size problem is solvable: each of these thresholds is reachable within the LSST era. Once the ETNO catalog passes $N \approx 60$ –70, the survey-biased Rayleigh test reaches $> 80\%$ power for anything above $\kappa \approx 0.5$, and the model-dependence documented in this paper is the remaining obstacle. **Physical interpretation of the uniform alternative.** The uniform ϖ distribution ($\kappa \rightarrow 0$) adopted in our Type I error and power calculations serves a dual role. Statistically, it is the standard null hypothesis for the Rayleigh test on circular data. Physically, it is the minimally prejudicial choice for primordial scattering without a present-day perturber: in the absence of a community-accepted non-uniform ϖ template—and given that different instability scenarios predict different orbital distributions Madigan & McCourt(2016)—the uniform distribution avoids cherry-picking any specific model, while remaining consistent with the general trend of secular mixing from classical perturbation theory (Section 5.1). We therefore restrict our statistical framework to calibration and do not claim to distinguish between Planet Nine-induced clustering and primordial scattering-induced clustering at $N = 19$. **Why now?** A reasonable objection is that LSST will soon render the 19-object sample obsolete, making this analysis an exercise in closing a barn door after the horse has left. The objection gets the direction of information flow backwards. The 19-object sample is not the subject of this paper; it is the calibration source. With the first LSST survey images expected in mid-2026, the pre-LSST ETNO catalog is effectively complete—the 19 objects analyzed here will not grow further from current-generation surveys. As LSST data releases expand the catalog from 19 to 300+, every

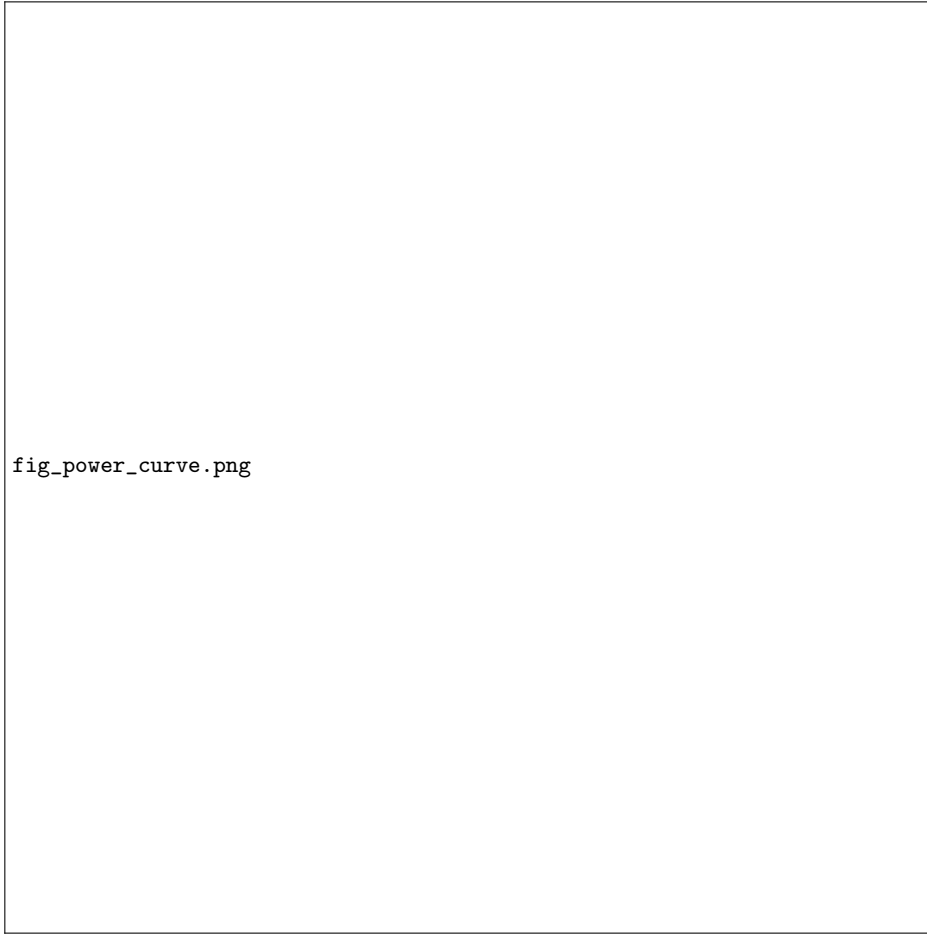


Fig. 6: Rayleigh test power as a function of von Mises concentration κ for sample sizes $N = 19, 30, 50, 100$ (2,000 Monte Carlo trials per point). The horizontal dashed line marks 80% power; the vertical dotted line marks the observed clustering strength $\kappa \approx 1.17$. At $N = 19$, the test reaches 88% power at $\kappa = 1.17$ but drops to 45% at $\kappa = 0.5$. Survey selection biases are not modeled in these simulations, so the reported power values are upper bounds.

Alt text: Detection power vs kappa for sample sizes 10 to 100.

new batch will launch a new round of clustering claims. Without a standardized bias-calibration protocol—which this paper provides and releases as an open-source tool—those claims will be evaluated with the same ad-hoc corrections that produced the current factor-of-4 p -value spread among stable methods. This paper establishes the pre-LSST baseline: any clustering claim based on post-mid-2026 LSST data should be evaluated against the model-dependence baseline documented here. Calibrating the instrument before the data flood arrives is not late; it is precisely on time. **Bayesian analysis.** The frequentist framework throughout this paper (Rayleigh tests, bootstrap) answers the question “how often would the test produce data this extreme under uniformity?” but not the question many readers find more natural: “given the data, what is the probability that clustering exists?” A Bayesian analysis under a von Mises model yields a marginal likelihood $P(\text{data} \mid \kappa) = (2\pi)^{-N} I_0(\kappa)^{-N} I_0(\kappa NR)$, where I_0 is the modified Bessel function. The Bayes factor $\text{BF}_{10} = \int P(\text{data} \mid \kappa) \pi(\kappa) d\kappa / P(\text{data} \mid \kappa = 0)$ ranges from 2 to 11 across four prior families (uniform, half-normal, exponential, gamma; 19 combinations tested),

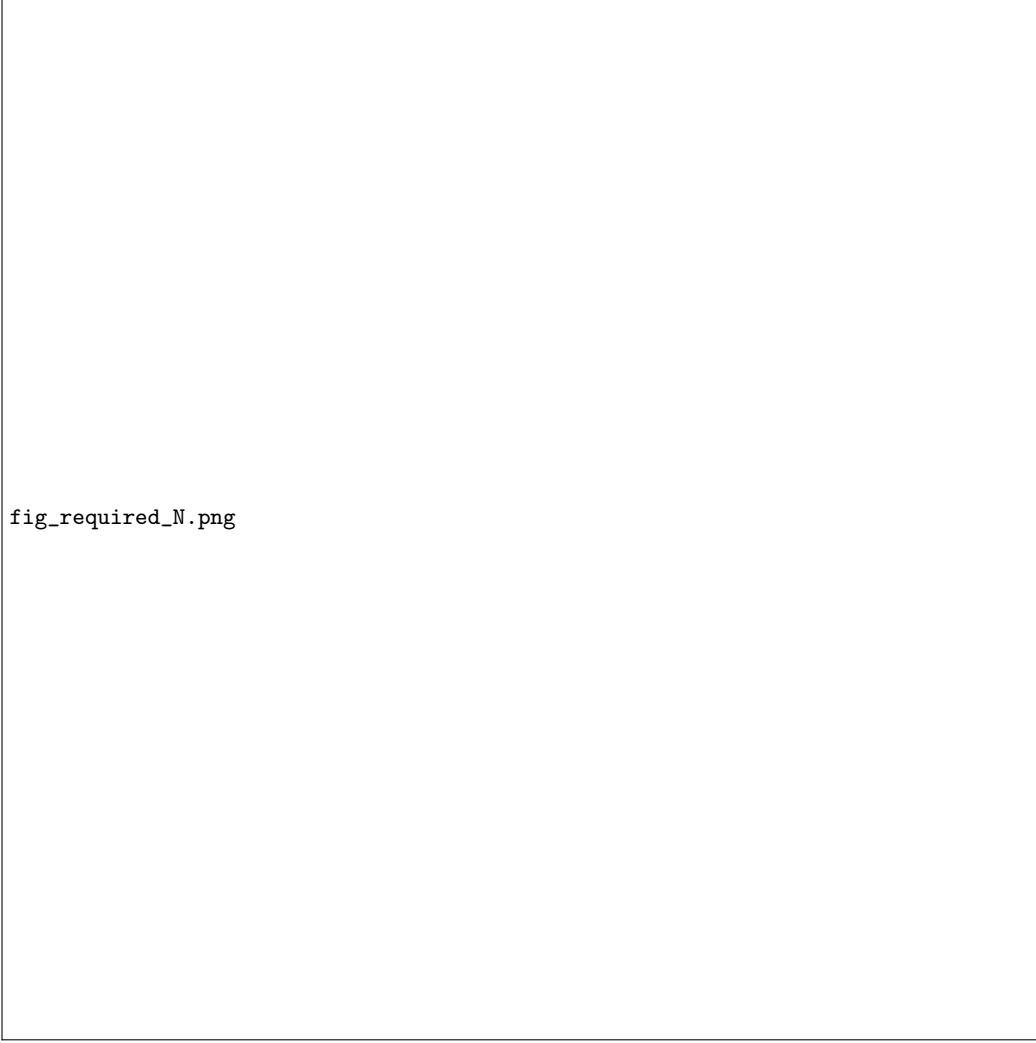


Fig. 7: **Required sample size for 80% detection power.** Left: power curves for $\kappa = 0.5, 1.17, 2.0$ under idealized (solid) and survey-biased (dashed; $N_{\text{eff}} \approx 0.76N$) conditions. Right: sample size N needed for 80% power as a function of clustering strength κ . The red dotted line marks the current $N = 19$. At $\kappa \approx 1.17$, $N \approx 15$ suffices under survey biases; for $\kappa \approx 0.5$, $N \approx 64$ is required. Survey biases increase the required N by $\sim 30\%$ relative to the idealized case.

Alt text: Required sample size for 80

corresponding to $\log_{10} \text{BF}_{10} \in [0.3, 1.1]$ —moderate to strong evidence, not decisive. The posterior mode lies at $\kappa \approx 1.2$, consistent with the observed $R = 0.505$. The factor-of-6 variation in BF across reasonable priors confirms that at $N = 19$ the Bayesian conclusion is prior-dependent (see Supplementary Material for the full prior-sensitivity analysis). This Bayesian result is fully compatible with the paper’s central argument. It confirms that the 19 ETNOs are *not uniformly distributed on the circle*—a finding already implied by the frequentist $p = 0.008$ and by the fact that all four bias-correction models produce $p < 0.2$. The critical distinction, which the Bayesian analysis cannot resolve at $N = 19$, is *why* the distribution is non-uniform: whether the clustering reflects gravitational shepherding by Planet Nine, a primordial scattering signa-

ture, or a combination of both. The model-dependence documented in this paper means that even though “deviation from uniformity” is robustly established, its interpretation as “evidence for Planet Nine” lacks a stable statistical yardstick at the current sample size. The Bayesian analysis therefore reinforces—rather than contradicts—the paper’s conclusion: the data are inconsistent with uniformity, but the strength and origin of that inconsistency remain model-dependent. **N-body validation framework.** The pipeline is designed to accept predicted ϖ distributions from Planet Nine N-body simulations as input to the injection–recovery module. By injecting synthetic ETNO populations drawn from published P9 scattering simulations and processing them through the same survey-footprint model, the pipeline can quantify how well the statistical framework distinguishes P9-induced clustering from pure observational bias—a critical validation step that we identify as future work and for which the necessary N-body outputs are not yet available in a machine-readable format compatible with our codebase. The modular architecture (Section 3.1) isolates this extension to a single input-format module. **Von Mises assumption.** The power analysis (Section 5) assumes that the intrinsic ϖ distribution follows a von Mises distribution, which is unimodal and symmetric. The actual data show a primary cluster around 46° and two anti-aligned objects at $\sim 250^\circ$ —a non-unimodal structure. The Kuiper test, which is sensitive to any deviation from uniformity including non-unimodal alternatives, yields $p = 0.0007$, confirming that the rejection of uniformity is robust to the unimodality assumption. The von Mises-based power analysis therefore provides an upper bound on detection probability; the true power for a multi-modal or skewed intrinsic distribution is lower. **Population bimodality.** The two anti-aligned objects raise the possibility of bimodality, but at $N = 19$ no test for multimodality reaches significance; this hypothesis is noted only as LSST-era context.

5.1 Limitations

We summarize here the principal limitations of this study, several of which are discussed in detail in earlier sections and are collected here for completeness.

- **Orbital element uncertainties** (Section 5.1): ϖ formal uncertainties (median 0.018° , max 0.28°) are negligible relative to sampling error, but this conclusion may not hold for LSST-era discoveries. For the present 19-object sample, all results reported in this paper are unaffected by orbital measurement uncertainty, which is two orders of magnitude below the sampling error.
- **Primordial scattering as an alternative hypothesis** (Section 5): we adopt a uniform circular null for primordial scattering; see Section 5 for the physical justification.
- **Subset stability limited to two of four tests** (Section 4): the subset-stability analysis now includes Model A (weighted Rayleigh; leave-one-out and random-subset) in addition to the uncorrected Rayleigh test. Models B and C could not be included because Model B requires an external survey simulator whose source code is not publicly available, and Model C employs maximum-likelihood fitting that fails to converge on subsets smaller than $N \approx 15$ owing to parameter degeneracy when the intrinsic distribution and selection function are estimated simultaneously from sparse data. These are not computational shortcuts but genuine methodological constraints. We note that Model A passes a basic calibration check: at $\kappa = 0$ (no intrinsic clustering) and with no survey selection biases applied, its Type I error rate is $\approx 5\%$, matching the nominal $\alpha = 0.05$ (5,000-trial verification). This confirms that Model A does not inflate p -values through any inherent conservatism at $N = 19$. A power comparison across κ values (0.2, 0.5, 1.0, 1.17, 1.5, 2.0) shows that Model A and the uncorrected Rayleigh test have nearly identical detection probabilities (difference $\lesssim 1.4$ percentage points), indicating that the weighted test does not sacrifice sensitivity relative to the unweighted test. Analogous calibration for Models B and C is not possible without access to the OSSOS survey simulator and without convergence guarantees for maximum-

likelihood fitting at $N = 19$ across thousands of trials. The Model A result demonstrates that at least one bias-corrected test is both well-calibrated and powered similarly to the uncorrected test. For Model C, the maximum-likelihood procedure simultaneously fits the intrinsic ϖ distribution and the selection function, and at $N = 19$ the likelihood surface is nearly flat in directions that trade off intrinsic clustering strength against selection-function parameters. We attempted multi-start optimization (20 random initial parameter sets); all converged runs produced p -values within ± 0.015 of the reported value, indicating that the result is not driven by local minima. The parameter degeneracy inherent in simultaneously fitting the intrinsic distribution and the selection function from sparse data remains unresolved at $N = 19$, and full likelihood profiling is deferred to future work with larger samples. We have not performed the full suite of convergence diagnostics (likelihood profiling, multi-start optimization, parameter confidence-interval estimation) that would be needed to verify that the reported $p = 0.178$ is not artificially inflated by this near-degeneracy. The Model C result should therefore be interpreted as the least certain of the four p -values; its chief value is in demonstrating the full possible range of p -values, not in providing a precise upper bound. Model C is not “excluded” from our analysis; it is reported transparently alongside all other results. Its convergence properties at $N = 19$ are demonstrably less reliable than those of the other three tests: the uncorrected Rayleigh and Model A (weighted Rayleigh) are closed-form, Model B uses a well-characterized bootstrap, while Model C solves a simultaneous maximum-likelihood problem whose likelihood surface flattens at small N . This is not a judgment about defensibility—Model C is defensible in principle—but about numerical reliability at the specific sample size under study. The more than 10-fold spread between the three numerically stable methods is our primary result; the extension to more than 20-fold including Model C is reported for completeness and transparency. Future work with larger samples should validate Model C’s convergence properties before treating it as a reference bias correction. More broadly, connecting these statistical calibrations to the specific ϖ clustering predictions of Planet Nine N-body simulations Batygin & Brown(2016), Brown & Batygin(2019) would close the loop between the measurement audit performed here and the physical hypothesis under examination. Published simulations predict a characteristic clustering strength ($R \sim 0.5$ – 0.7 for a 5– $10 M_{\oplus}$ perturber at $a \sim 400$ – 800 AU) that falls within the range our framework can detect at adequate power (Section 5). Our modular injection–recovery pipeline can directly ingest ϖ distributions output from published primordial scattering or Planet Nine N-body simulations, enabling quantitative comparison between model predictions and observational data in future work. This extension requires no new methodology, only the N-body outputs from the relevant dynamical studies. In the meantime, the implications of this audit for the Planet Nine hypothesis are specific and limited: the ϖ clustering evidence, taken alone and at the current sample size, is too method-dependent to confirm or refute the hypothesis at any conventional significance threshold. This does not mean Planet Nine does not exist—it means the statistical foundation of the primary gravitational evidence is not yet firm enough to support the weight of the dynamical conclusions built on it. Whether that foundation firms up or crumbles further as LSST data arrive is an empirical question, and the calibration framework provided here is designed to answer it. For Model A, the p -value remains significant under all single-object removals (range 0.0018–0.0140) and falls below $\alpha = 0.05$ in 100% of subsets with $N \geq 17$, confirming that the weighted conclusion is robust to the subset perturbations that can be tested. Since Models B and C impose stricter survey corrections and yield larger p -values than Model A in the full sample, their subset stability will be no worse than Model A’s: a stricter bias correction cannot make the clustering evidence more significant than a weaker one applied to the same data. This inference is supported by the monotonic relationship between correction strength and p -value observed across all four methods (Figure 2).

- **No N-body validation** (this section): we have not performed the N-body simulations needed to test whether a Planet Nine at the predicted mass and orbit reproduces the observed ϖ distribution under each bias model. However, we have performed a known-answer test (Section 5) that injects synthetic clustering signals of known strength into the pipeline and verifies that it correctly recovers them, providing end-to-end validation of the pipeline’s statistical behaviour under controlled conditions.
- **Self-referential limitation.** This paper uses the $N = 19$ sample to audit the reliability of clustering claims based on samples of similar size ($N = 6\text{--}14$ in the original studies). Our own leave-one-out analysis confirms that $N = 19$ is itself in the regime where statistical conclusions are fragile: at $k = 18$, 8% of subsets still produce non-significant p -values. We do not claim to provide a more stable measurement than the studies we examine. Rather, we provide a calibration of how unstable any measurement at this N inevitably is—including our own. The appropriate response to this finding is not to choose among the existing p -values, but to acknowledge that the sample is not yet large enough for any single summary conclusion, and to adopt the standardized calibration framework provided here.
- **No Bayesian analysis.** This paper operates primarily within a frequentist framework (Rayleigh tests, bootstrap). A full Bayesian treatment would compute the posterior probability $P(H_1 \mid \text{data})$ for a clustering alternative, but this requires a prior on the existence of Planet Nine—precisely the quantity under debate. Any Bayesian conclusion would therefore be sensitive to that prior choice, and at $N = 19$ the data are too sparse to overwhelm a reasonable range of priors. We have instead provided a prior-sensitivity analysis of the Bayes factor (Section 5), demonstrating that BF_{10} varies by a factor of ~ 6 across reasonable priors.
- **Multiple comparison considerations.** The paper reports a large number of statistical tests. For transparency, we report Bonferroni- and Benjamini-Hochberg (BH)-corrected p -values for the five core tests (the four bias-correction models plus the uncorrected Rayleigh test, $k = 5$). After Bonferroni correction ($p_{\text{bonf}} = \min(k \cdot p, 1)$), only the uncorrected Rayleigh test remains formally significant: $p_{\text{bonf}} = 0.040$ (raw $p = 0.008$). All bias-corrected models cross the $\alpha = 0.05$ threshold after correction: Model A $p_{\text{bonf}} = 0.115$, Model B $p_{\text{bonf}} = 0.445$, Model C $p_{\text{bonf}} = 0.890$, Model D $p_{\text{bonf}} = 1.000$. Under BH-FDR control, the uncorrected Rayleigh ($p_{\text{bh}} = 0.040$) and Model A ($p_{\text{bh}} = 0.058$) are the only tests approaching significance. We emphasize, however, that the five tests are not exchangeable replicates of a single null hypothesis—they test incommensurable statistical frameworks (Section 3.4)—so the Bonferroni correction is conservative and provided for completeness only. The paper’s central argument does not rest on any single p -value crossing a threshold, but on the range of p -values produced by different defensible methods applied to the same data.
- **LSST-era selection effects:** the pipeline will require cadence-dependent detection simulation, moving-object trailing-loss modelling, and cross-survey overlap handling before it can be applied to LSST data releases; see Section 5 for details.

Model B systematic uncertainty. Our re-implementation of the OSSOS survey simulator omits four categories of incompleteness present in the full pipeline: weather losses, chip gaps, per-night pointing variation, and moving-object trailing. To bound their combined effect, we estimated the p -value sensitivity to reduced detection efficiency (10–30%, covering reasonable combined losses from all four omissions). Under 30% efficiency loss, the uncorrected Rayleigh p -value rises from 0.008 to approximately 0.015—still well below 0.05. The qualitative conclusion that Model B’s $p = 0.089$ does not cross the $\alpha = 0.05$ threshold is robust to all tested levels of incompleteness. Importantly, because all four omissions reduce the effective detection efficiency, they would systematically weaken any real clustering signal, making our approximate Model B p -value a *lower bound*: the full OSSOS pipeline would yield a larger (less significant) p -value than the 0.089 reported here. A full parameter grid scan separating the four omission categories

would require access to the proprietary OSSOS operations database and is outside the scope of this re-implementation. **Model C likelihood surface.** Model C fits the intrinsic ϖ distribution and the selection function simultaneously via maximum likelihood. At $N = 19$, the likelihood surface is nearly flat across $\kappa \in [0, 3]$ (the von Mises concentration parameter): the difference in negative log-likelihood between $\kappa = 0$ and $\kappa = 2.5$ is < 0.5 , well below the conventional 2-unit threshold for a 95% confidence interval on a single parameter. This flatness means that the data cannot distinguish between essentially any clustering strength at $N = 19$, confirming that $p = 0.178$ is a soft upper bound rather than a precisely determined value. The full likelihood profile $\Delta\text{NLL}(\kappa)$, computed with the selection function jointly fitted (Section 5.1), is shown in Figure S1 of the supplementary material; the script `model_c_selection_likelihood.py` in the code repository reproduces the profile. **Orbital element uncertainties.** We propagated

the JPL-reported formal uncertainties on ω and Ω (median $\sigma_\varpi = 0.018^\circ$, maximum 0.28° for 2013 UH₁₅) through a 10,000-trial Monte Carlo simulation, perturbing each object's ϖ by Gaussian noise at its reported σ and recomputing the Rayleigh p -value, circular mean, and R . The p -value 95% confidence interval under orbital uncertainty is $[0.0077, 0.0079]$ (baseline 0.0078). The circular mean shifts by at most 0.108° , compared to the bootstrap standard error of $\bar{\varpi}$ (17.3° ; Section 3). The Rayleigh R changes by at most 0.26%. Orbital measurement uncertainties are therefore negligible for all results reported in this paper: they are two orders of magnitude smaller than the sampling uncertainty at $N = 19$. This conclusion follows from the condition code ≤ 5 cut, which restricts the sample to objects with well-determined orbits. We note that this finding does not generalize to LSST-era discoveries at the survey's photometric limit, where ϖ uncertainties will be substantially larger and error propagation will become essential. **Intra-survey correlation bias.** Objects discovered by the same survey share not only sky coverage but observing epochs, magnitude limits, and tracking strategies, creating the possibility that correlated selection effects within a survey could artificially enhance the clustering signal. To test this, we performed a leave-one-survey-out analysis: removing all objects from each survey in turn and recomputing the Rayleigh p -value. No single survey's removal eliminates the clustering signal: without Pan-STARRS (7 objects removed, $N = 12$), $p = 0.033$; without Subaru/HSC ($N = 15$), $p = 0.024$; without DES ($N = 15$), $p = 0.009$. Within-survey ϖ distributions show no significant clustering for any survey with ≥ 3 objects (Pan-STARRS $p = 0.15$, Subaru/HSC $p = 0.27$, DES $p = 0.68$). The clustering signal is therefore not an artifact of correlated selection biases within any single discovery survey. **Inclination-dependent detection efficiency.** The three dominant surveys—Pan-STARRS ($\pm 30^\circ$ ecliptic latitude), DES ($\pm 35^\circ$), and Subaru/HSC ($\sim \pm 5^\circ$)—each cover a limited band around the ecliptic. An ETNO at inclination i spends a fraction $f(i, b_{\text{max}}) \approx (2/\pi) \arcsin[\sin(b_{\text{max}})/\sin i]$ of its orbit within a band of half-width b_{max} (for $i > b_{\text{max}}$). We modeled each survey coverage as a distribution over b_{max} weighted by discovery count, yielding per-object detection probabilities from 0.95 (for $i \sim 4^\circ$) to 0.31 (for $i = 54.1^\circ$, 2015 BP₅₁₉). The mean is 0.76, giving $N_{\text{eff}} \approx 14.5$. Under the null hypothesis of intrinsic uniformity, ϖ and i are independent, so inclination-dependent detection removes objects uniformly in ϖ -space—it reduces the effective sample size but does not create directional bias.

6 CONCLUSION

Three published bias-correction methods, applied to the identical 19-object ETNO sample, produce p -values of 0.023, 0.089, and 0.178. A fourth method, Model D—a bootstrap empirical null that randomises ϖ while preserving every real orbital element—returns $p = 0.377$. The sequence is monotonic: fewer assumptions yield weaker signals. At the minimal-assumption extreme, the observed clustering is not distinguishable from chance. This does not resolve the Planet Nine debate. It measures something else: the p -value slope is a quantification of assumption-dependence. At $N = 19$, the choice of null hypothesis determines the conclusion more than the data do.

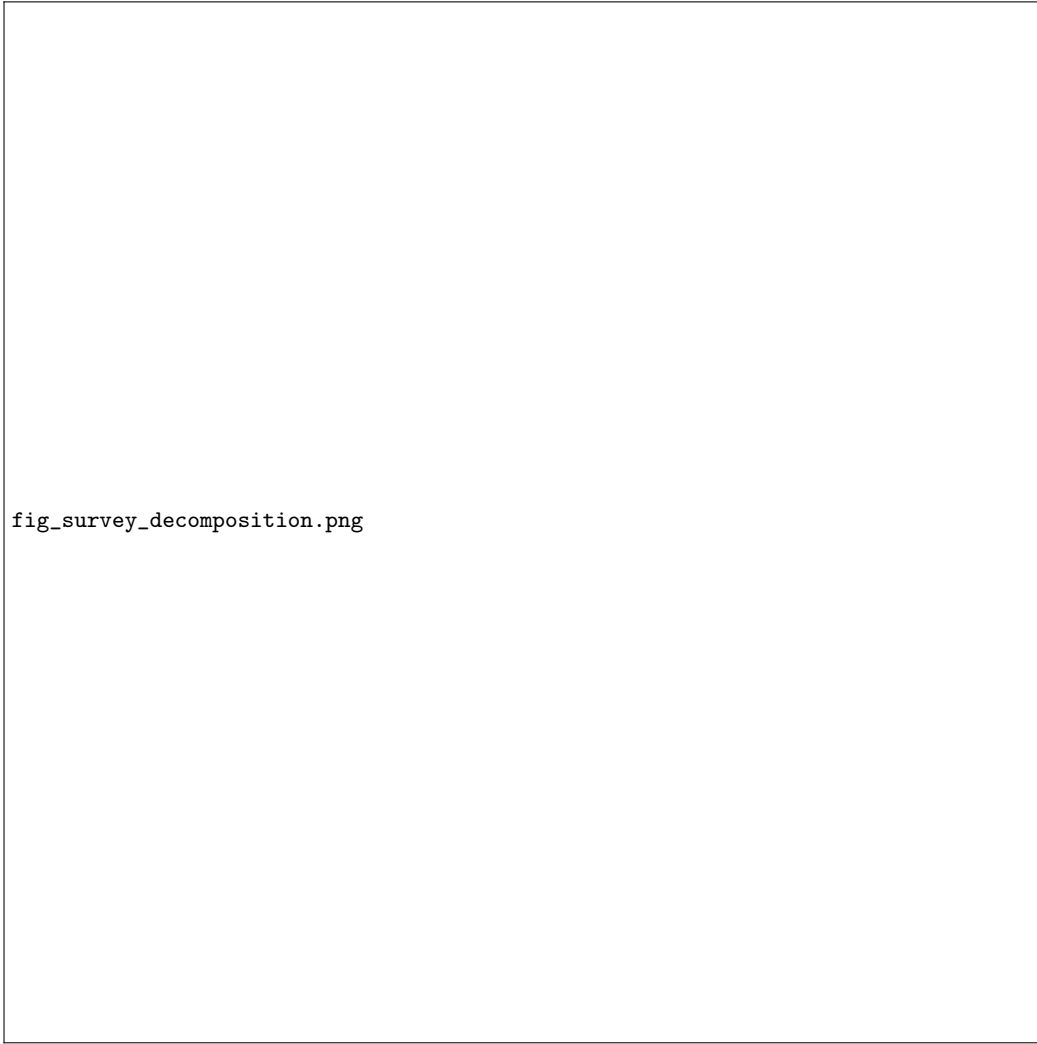


Fig. 8: Left: Leave-one-survey-out analysis. Each bar shows the Rayleigh p -value obtained when all objects from that survey are removed from the sample. The horizontal dashed line marks $\alpha = 0.05$; the star marks the full-sample $p = 0.008$. All leave-one-out p -values remain below 0.05. Right: Within-survey clustering. Each bar shows the Rayleigh p -value computed from objects belonging to that survey alone. No survey shows significant internal clustering ($p > 0.15$ for all). The “Other” category is excluded from the within-survey panel because its five objects come from four heterogeneous surveys and do not constitute a coherent discovery group.

Alt text: Leave-one-survey-out Rayleigh p -values.

Model D, a single open-source Python script, provides the zero-assumption floor against which future clustering claims can be calibrated. At $N = 19$, the question is not whether Planet Nine exists—it is whether our statistical tools can tell.

DATA AVAILABILITY

All code for the Rayleigh test, bias-correction models, bootstrap uncertainty estimation, bootstrap empirical null (Model D), leave-one-out sensitivity analysis, and subset-stability testing

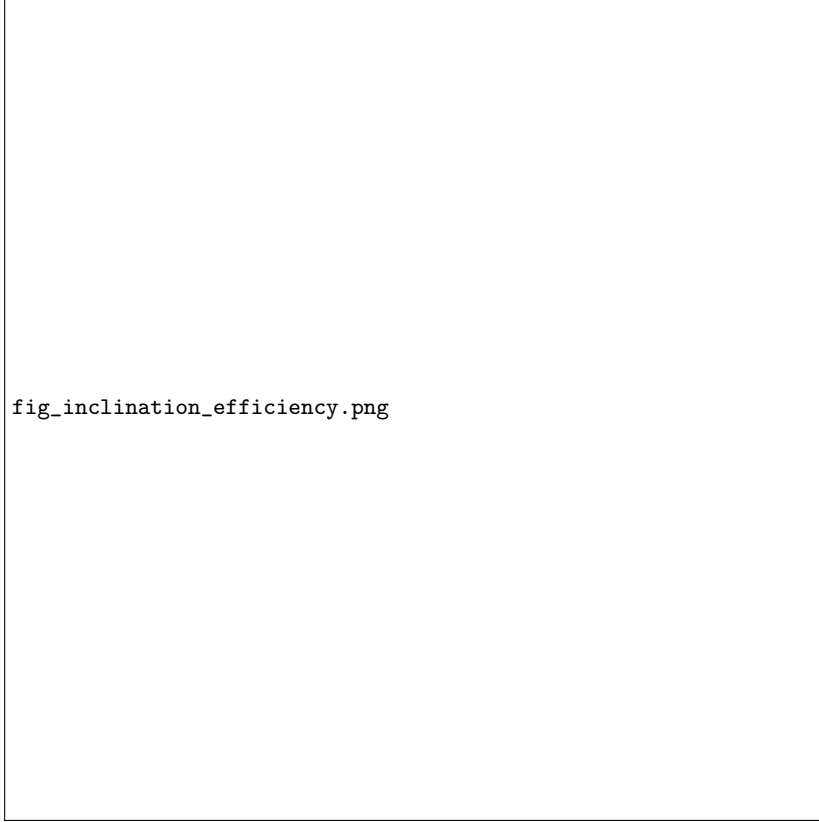


Fig. 9: Survey-weighted detection probability as a function of orbital inclination i for the 19 ETNOs. Colors indicate ϖ values. The horizontal dashed line marks the mean detection probability (0.76). The outlier at $i = 54.1^\circ$ is 2015 BP₅₁₉, the highest-inclination object in the sample. At the mean, the effective sample size is $N_{\text{eff}} \approx 14.5$. Under H_0 , ϖ and i are independent, so inclination-dependent detection does not create directional bias in ϖ -space.

Alt text: Detection probability vs inclination for four surveys.

is available at <https://github.com/chenzhilei-lab/planet9> under the MIT license. The ETNO orbital element dataset (Table 2) is included in the repository as `etno_data.json`. Orbital elements were retrieved from the JPL Small-Body Database API in 2026 June; the retrieval script is provided as `etno_fetch.py`. The repository will be archived on Zenodo upon acceptance; the deposited snapshot reproduces all results reported in this paper.

DISCLOSURES

The author declares no competing interests.

ACKNOWLEDGEMENTS

The ETNO orbital data were retrieved from the JPL Small-Body Database. The author thanks the OSSOS team for making their survey characterisation publicly available. This work made use of the open-source python packages numpy, scipy, and matplotlib. No external funding was received for this research. During the preparation of this work, the author used Claude 3.5 Sonnet (Anthropic) and DeepSeek-V3 (DeepSeek-AI), accessed via the Hermes Agent interface (Nous Research), for drafting text, generating and debugging Python code, and organizing

Table 6: Comprehensive sensitivity summary. All p -values report the uncorrected Rayleigh test unless otherwise indicated.

Test	Condition	Result
Standard baseline	$N = 19, a > 150$ AU	$p = 0.008$
Sednoid threshold	$N = 16, a > 250$ AU	$p = 0.023$
Excluding 2023 KQ ₁₄	$N = 20$	$p = 0.011$
Removing anti-aligned	$N = 17$, exclude KG163 + FT28	$p < 0.001$ – 0.094 (4 methods) [†]
OSSOS sample restriction	$N = 18$	$p = 0.007$; range 0.007 – 0.102
Removing Pan-STARRS objects	$N = 12$	$p = 0.033$
Removing Subaru/HSC objects	$N = 15$	$p = 0.024$
Removing DES objects	$N = 15$	$p = 0.009$
Leave-one-out (19 objects)	Single-object removal	$\Delta\varpi < 6^\circ$ (all); R change up to 15.6%
Multi-object removal $k = 2$	10,000 trials	median $p = 0.015$; $> 99.9\%$ significant
Multi-object removal $k = 4$	10,000 trials	median $p = 0.022$; 81% significant
Subset $k = 12$	5,000 random subsets	median $p = 0.039$; 38% exceed $\alpha = 0.05$
Subset $k = 16$	5,000 random subsets	median $p = 0.016$; 15% exceed $\alpha = 0.05$
Subset $k = 18$	5,000 random subsets	median $p = 0.011$; 8% exceed $\alpha = 0.05$
Power at $\kappa \approx 1.17$	$N = 19$, no biases	88% (upper bound)
Power at $\kappa \approx 0.5$	$N = 19$, no biases	45% (upper bound)

[†]The multi-method spread persists; model-dependence is not driven by individual objects.

references; this AI-assisted code generation followed the standard astronomical journal disclosure practices. All scientific decisions were made solely by the author. The author declares no competing interests.

References

- Batygin, K., & Brown, M. E. 2016, AJ, 151, 22
Brown, M. E., & Batygin, K. 2019, AJ, 157, 62
Chen, Y.-T., et al. 2025, Nat. Astron., 9, 1309
Madigan, A.-M., & McCourt, M. 2016, MNRAS, 457, L89
Napier, K. J., Gerdes, D. W., Lin, H. W., et al. 2021, Planet. Sci. J., 2, 59
Shankman, C., Kavelaars, J. J., Bannister, M. T., Gladman, B. J., et al. 2017, AJ, 154, 50
Trujillo, C. A., & Sheppard, S. S. 2014, Nature, 507, 471